Title:

Harnessing carbon tax for embodied emissions reduction: A case study of Australian shopping centres

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ABSTRACT

Efforts to reduce operational emissions in the built environment are now shifting towards achieving embodied emissions reductions, which contribute to approximately 20% of Australia's greenhouse gas emissions. Strategies like policy mandates, carbon pricing, and industry awareness programs target these embodied emissions, with a carbon tax gaining popularity for its effectiveness. Despite its proven effectiveness, Australia is yet to reinstate a carbon tax as an emissions reduction mechanism since it was repealed in 2014.

This study utilises python based parametric modelling to demonstrate how a carbon tax influence material selection in construction. Analysing over 8,820 scenarios in an Australian shopping centre case study, the model establishes significant potential reductions in embodied emissions (up to 59%) and cost savings (up to 47%) compared to the business-as-usual scenarios of different shop types. The findings further reinforce that when a carbon tax is incorporated, the lowest cost assembly combination, is similar to the optimal assembly combination achieving emissions and cost reductions equally and assembly combinations could lead to 51% emissions decrease and 46% cost savings.

The study emphasises the importance of re-implementation of the carbon tax approach to achieve sustainability targets and reduce embodied emissions in the built environment.

Keywords: Australia, Carbon tax, Embodied GHG emissions, Life cycle cost, Shopping centres

1 INTRODUCTION

Regulations and policies in many countries mandate or encourage the use of environmentally sensitive building materials and assemblies as a means to achieve emissions reduction objectives (Curmi et al., 2022; Skillington et al., 2022; World Bank, 2022). Various strategies are employed, including a combination of taxes, subsidies, policy mandates, and revisions to building standards and requirements. Among these approaches, carbon pricing stands out as a promising global strategy for mitigating the adverse environmental impacts of the construction industry (Andersson, 2019; Dominioni, 2022; Gorbach et al., 2022; Nakano & Washizu, 2022). Specifically, the implementation of a carbon tax has become widespread within the construction sector worldwide, to encourage the selection of building materials with lower embodied emissions (Huang et al., 2018; Laes et al., 2018; Mehrzad et al., 2023; Sathre & Gustavsson, 2007; Shipworth, 2002). The carbon tax mechanism operates on a pricing basis, assigning a fixed price to greenhouse gas (GHG) emissions (Sathre & Gustavsson, 2007).

Applying a carbon tax to the construction process and building materials can have immediate and tangible effects, primarily due to the associated increase in costs (Mehrzad et al., 2023; Mollaei et al., 2023). The financial implications of this tax are likely to induce a shift in behaviour among developers and contractors, prompting them to seek alternative construction practices and materials with lower embodied emissions (Abeydeera et al., 2019; López et al., 2015; Shipworth, 2002). Consequently, the enforcement of a carbon tax is widely recognised as one of the most effective strategies for promoting environmental performance within the construction industry and has the potential to steer the industry towards eventual carbon neutrality (Andersson, 2019; Metcalf, 2019; Röck et al., 2020; Ropo et al., 2023).

An illustrative case study is Australia, where the introduction of a carbon tax in 2012 led to a significant reduction in greenhouse gas (GHG) emissions within the construction sector (Crowley, 2017; Hammerle et al., 2021; O'Gorman & Jotzo, 2014). However, it is important to note that the tax was later repealed in 2014 (Crowley, 2017; Wong & Zapantis, 2013). Subsequent evidence indicates that GHG emissions experienced a resurgence after the repeal of the tax (World Bank, 2022). These findings emphasise the importance of a consistent and long-term implementation of carbon taxation measures to achieve a sustained reduction of emissions.

Given that buildings are the largest users of energy and emitters of greenhouse gas emissions on the demand side, substantial progress has been made in implementing demand-side mechanisms aimed at reducing energy use and emissions generation (Fenner et al., 2018; Ibn-Mohammed et al., 2013; Ohene et al., 2022). Implementing energy-efficient technologies, adopting renewable energy sources, and promoting energy conservation practices have all played significant roles in achieving considerable reductions in energy demand within the construction industry (Laes et al., 2018; Stephan & Stephan, 2020). This highlights the significance of demand-side strategies in mitigating the environmental effects of buildings and advancing sustainability goals. Therefore, it is crucial to evaluate the implications of a carbon tax on building material selection as it may affect behavioural changes in stakeholders (Ratanakuakangwan & Morita, 2022).

This research investigates the effect of enforcing a carbon tax on material selection of Australian shopping centres as a building form. Shopping centres are experiencing rapid growth in Australia, primarily due to population growth and urban sprawl (Weththasinghe et al., 2020). Compared to other commercial properties, retail shops in shopping centres undergo exceptional refurbishments every 2 to 10 years. As a result of the frequent refurbishments (typically between two to ten years), building materials in shopping centres are prematurely replaced due to economic, functional, and social obsolescence. This excessive use of resources leads to increased embodied GHG emissions (Weththasinghe et al., 2022a). However, despite these adverse effects, there is a lack of understanding regarding the environmental effects of shopping centres in Australia, which hinders progress towards achieving sustainability. Hence, this research simulates different scenarios of carbon taxes as an approach to mitigate embodied emissions of Australian shopping centres through material selection and proposes potential cost and emissions reductions.

Although the costs associated with sustainable material selection is acknowledged as a hindrance, the imposition of taxes and levies is recognised as a motivating factor (Pomponi & Moncaster, 2016; Quesada-Pineda et al., 2018; Skillington et al., 2022). Carbon pricing, among various taxation methods, has been established as a successful mechanism globally for promoting sustainability within the built environment (Curmi et al., 2022; Dominioni, 2022; Green, 2021b).

The adoption of a comprehensive policy on carbon pricing is crucial for achieving significant reductions in embodied emissions within the built environment sector. By aligning economic incentives with environmental goals, carbon pricing can drive market transformation, encourage sustainable practices, and redirect investments towards low-carbon solutions. However, it is crucial to consider equity and social implications to ensure a fair and just transition. As the urgency to combat climate change intensifies, policymakers must prioritise implementing robust carbon pricing policies to foster sustainable development in the built environment.

2 BACKGROUND

The emission of greenhouse gases (GHGs) is widely acknowledged as a significant environmental concern in the context of the built environment (Allen et al., 2022; Huang et al., 2022; Röck et al., 2020; Yu et al., 2020). In addition to GHG emissions resulting from the operational phase of buildings, there is also a substantial impact from embodied GHG emissions associated with construction materials and refurbishments (Dixit, 2017; Ibn-Mohammed et al., 2013; Röck et al., 2020; Ropo et al., 2023; Shipworth, 2002). Governments worldwide have implemented various strategies to address and mitigate both operational and embodied GHG emissions in the construction and use of buildings, aiming to achieve sustainability goals (Holloway, 2021; Meng & Yu, 2023; Skillington et al., 2022).

Among these strategies, carbon pricing has emerged as a promising approach to tackle the adverse environmental impacts of the construction industry (Andersson, 2019; Berthe et al., 2023; Boyce, 2018; Rontard & Hernandez, 2022). Carbon pricing serves as a tool to discourage GHG emissions and promote more sustainable practices. The landscape of carbon pricing mechanisms is continuously evolving, with a notable increase in adoption. As of 2019, 46 national and 28 subnational jurisdictions had implemented carbon pricing systems (World Bank, 2022).

Six distinct types of carbon pricing mechanisms have been identified. These include internal carbon pricing, emissions reduction credit schemes, emissions trading systems, hybrid schemes, carbon taxes, and command and control mechanisms. Each mechanism operates with its own set of regulations and approaches, providing a range of options for governments and organisations to address GHG emissions and promote sustainable practices in the built environment (Baranzini et al., 2017; Narassimhan et al., 2018).

The implementation of the carbon tax mechanism is widely practised in the global construction industry with the aim of promoting the selection of building materials with lower embodied emissions. Various studies (Andersson, 2019; Laes et al., 2018; Shi et al., 2019; World Bank, 2022) have recognised the effectiveness of the carbon tax mechanism in influencing developers and contractors to opt for materials with reduced emissions content due to the associated cost increase.

The introduction of a carbon tax in construction processes and material selection can yield immediate effects through heightened costs (Abdi et al., 2018; Chen et al., 2015). As a result, developers and contractors are likely to alter their behaviour by shifting towards materials with lower embodied emissions as substitutes (Elaouzy & El Fadar, 2023; Luo et al., 2021; Mehrzad et al., 2023). Consequently, the carbon tax mechanism has been recognised as an effective tool for encouraging the increased use of materials and assemblies with lower embodied emissions. Furthermore, the cost of embodied GHG emissions over the life cycle of a building serves as a reliable indicator of its environmental sustainability (Allen et al., 2022; Röck et al., 2020; Stephan & Athanassiadis, 2017; Teh et al., 2019).

In order to determine the total carbon tax associated with the materials used in a building, it is necessary to quantify the amount of greenhouse gas (GHG) emissions embodied in the project. Various approaches can be employed to estimate the overall embodied GHG emissions attributable to building materials (Czachorski & Leslie, 2009; Finnegan et al., 2018; Lolli & Hestnes, 2014). One method involves directly converting embodied energy values into embodied GHG emissions values by applying a conversion factor (Dias & Pooliyadda, 2004; Graabak et al., 2014). In the context of Australia, the conversion factor for life cycle embodied GHG emissions is typically considered as 58.78 kg CO_2e/GJ (Langston et al., 2018). Another approach is to utilise carbon emission coefficients specific to different building materials, which express the amount of carbon emissions in kgCO_2e/kg of the respective materials (Dias & Pooliyadda, 2004; Hammond & Jones, 2008; Treloar et al., 2001). This approach is considered to be more reliable and accurate as it accounts for emissions on a material-by-material and assembly-by-assembly basis based on the quantities of materials employed. This method requires more resources and data, as carbon coefficients need to be developed for each building material (Bueno et al., 2016; Crawford et al., 2018; Gursel et al., 2014; Stephan et al., 2019).

Within the Australian context, the carbon tax mechanism was introduced as a policy measure and remained in effect for a span of two years, spanning from 2012 to 2014 (Burke, 2016). Notably, during the period of its enforcement, it was observed that emissions originating from the construction industry exhibited substantial reductions (Copland, 2020; Hammerle et al., 2021; Meng et al., 2013). This observation highlights the positive impact that the carbon tax policy had on curbing greenhouse gas emissions within this particular sector (Yu et al., 2017).

However, subsequent to the retraction of the carbon tax policy, there has been a concerning reversal in the emission trends, with levels surpassing those experienced prior to the policy's implementation (Yu et al., 2017). This shift suggests that the regulatory alteration, specifically the removal of the carbon tax, has contributed to an unfortunate increase in emissions beyond the baseline level (Copland, 2020; Hammerle et al., 2021; Meng et al., 2013). These findings illuminate the influential nature of regulatory changes in the environmental arena, highlighting that policy adjustments can yield profound consequences for emission patterns and overall environmental sustainability.

It is important to note that this pattern of emission fluctuations following regulatory changes can have far-reaching implications beyond Australia. The case of the carbon tax mechanism showcases the potential effectiveness of regulatory measures in reducing emissions, but also emphasises the risks associated with their discontinuation (Copland, 2020; Hammerle et al., 2021; Meng et al., 2013; Rontard & Hernandez, 2022). Such insights are of particular significance for policymakers, as they underscore the need for careful consideration and evaluation of environmental regulations to ensure long-term positive outcomes and mitigate adverse effects on climate change mitigation efforts (Baranzini et al., 2017; Green, 2021a; Gugler et al., 2023; Hao & Yang, 2022).

Australia, as a signatory to the Global Renewables and Energy Efficiency Pledge, has committed to enhancing renewable energy sources and improving energy efficiency as integral components of its climate action strategy (United Nations Climate Change, 2022). Concurrently, Australia recognises the critical role of the construction sector in achieving broader sustainability objectives (Australian Sustainable Built Environment Council, 2022). In response, Australia has articulated strategies to enhance sustainability in the construction sector, aligning with the goals of the Global Pledge to foster environmentally responsible building practices. This concerted effort stresses Australia's commitment to addressing both the energy and construction sectors as integral components of its comprehensive approach to global sustainability.

Consequently, the possibility of reintroducing a carbon tax arises; nevertheless, prior to its implementation, a comprehensive evaluation of the associated implications is necessary (Copland, 2020; Wong et al., 2019). Given that shopping centres in Australia represent a significant component of the built environment, they play a crucial role in mitigating emissions (Braslavsky et al., 2015). Therefore, it is of utmost importance to examine the potential repercussions of enforcing a carbon tax on the life cycle material costs associated with the development of shopping centres in Australia, as such measures have the potential to stimulate transformative changes. Lessons learned during this exercise also provide insights into the effect of a carbon tax on the material selection of other building types.

Overall, the implementation of carbon pricing mechanisms is a crucial step in achieving environmental sustainability goals, as it incentivises the reduction of GHG emissions in the construction industry and drives the adoption of more eco-friendly practices. The increasing number of jurisdictions adopting carbon pricing systems demonstrates the growing recognition of the importance of addressing GHG emissions within the built environment, and it signifies a global commitment to combat climate change.

3 METHODS

This study employed an inductive research strategy (Saunders et al., 2023) to address the initially stated research problem and gap. Moreover, a mixed research methodology incorporating case studies and a mathematical parametric model was utilised to quantify the cost of embodied greenhouse gas emissions of Australian shopping centres.

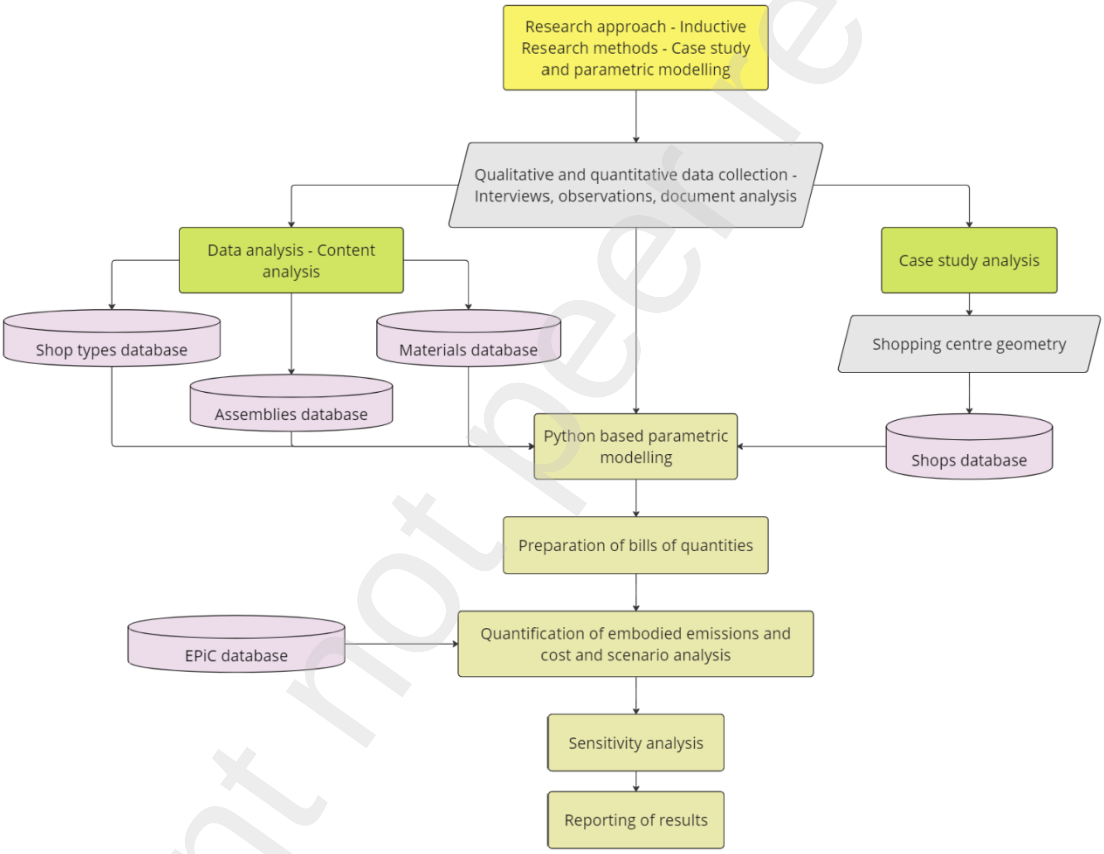

Figure 1: Research method

The case study method integrates both qualitative and quantitative data collection techniques, thereby proposing a mixed-method approach (Yin, 2017). Qualitative data was gathered through semi-structured interviews, document analysis, and on-site observations.

The interviews were conducted with professionals involved in shopping centre development, management bodies, and shopping centre managers themselves. Types of building materials and assemblies were directly inputted into the model as qualitative data. Additionally, quantitative data was obtained through interview findings, document analysis, and observations. Both qualitative and quantitative data were treated as quantitative data in mathematical modelling to obtain answers to the research questions.

A mathematical model developed in Python is employed in this study to serve the purpose of quantifying the LCEGHGE, LCMC and LCCEE associated with various combinations of building materials in the case study buildings. Given the substantial size of shopping centres and their utilisation of diverse building materials and assemblies, the calculation of LCEGHGE, LCMC and LCCEE for such structures becomes a complex task. The calculation process necessitates the inclusion of multiple data inputs concerning the materials, assemblies, quantities, and other data-intensive matrix calculations. The following sub sections demonstrate the research methodology in detail, elaborating on the parametric model development and case study selection for simulations.

3.1 Parametric model development

A crucial aspect of the model is its ability to perform scenario analysis, enabling rapid assessment and comparison of embodied GHG emissions and material costs for different assembly combinations. To meet this requirement, the model's architecture needed to exhibit resilience and flexibility. Consequently, the development approach for the mathematical model was carefully chosen, taking into account the identified requirements. In this regard, object-oriented programming was selected as the preferred modelling approach. While various programming paradigms in Python, such as imperative, functional, procedural, and object-oriented (Phillips, 2018; Python Software Foundation, 2023), can be adapted, the object-oriented paradigm was deemed the most suitable given its pragmatic nature and alignment with the aforementioned requirements.

This approach facilitated the identification of optimal combinations of building assemblies tailored to various shop types within shopping centres. The model's development unfolded in two key stages: (1) the establishment of databases and (2) the development of the computing core. In the computing core, four classes—*Materials*, *Assemblies*, *Shop*, and *Shoppingcentre*—were defined. The model leverages the methods and attributes associated with these classes to instantiate objects of *Materials*, *Assemblies*, *Shops*, and *Shoppingcentres*. These classes use data inputs recorded as quantitative data within five distinct databases, as shown in Table 1.

Table 1: Details of the databases defined in the model

Assemblies are categorised into eleven types, following elemental categories as defined by Australian Institute of Quantity Surveyors (2022): foundation, roof structure, columns, structural wall, internal wall, window, lintel, floor finish, wall finish, ceiling finish, and waterproofing. Over 60 construction assemblies are classified under these types. Additionally, the *Assemblies* database incorporates details on the materials and respective quantities constituting a unit of each assembly. The *Assemblies* database is openly accessible for reference (Weththasinghe et al., 2022b).

Information pertaining to various shop types within shopping centres was stored in the *Shops_catalogue* database. Within Australian shopping centres, a total of 13 distinct shop types were identified, encompassing anchor shops such as supermarkets and discount department stores and specialty shops such as those dedicated to clothing, cafes, restaurants, health and beauty, and services. These categories are delineated in accordance with the typical tenancy mix outlined by the Shopping Centre Council of Australia. A shopping centre is conventionally composed of two integral components: 1) the centre structure, which constitutes the core building with all structural elements (e.g., foundation, roof, columns, structural wall), and 2) the internal layout, encompassing the fit-out designs of anchor shops, specialty shops, and common areas. For modelling purposes, all these components are considered as distinct shop types. Specialty shops, for instance, are characterised by internal walls, floor finishes, wall finishes, and ceiling finishes, whereas anchor shops feature structural walls in place of internal walls, given their longer lifespan.

Shops in the *Shops* database are modelled using a 'shoe-box' approach, as presented in Figure 2. Accordingly, a shop is considered as a box with a length (l), width (w) and height (h), and in some cases, a span. These parameters are combined to generate bills of quantities.

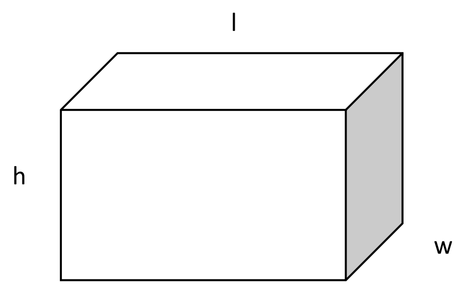

Figure 2: Shoe-box scenario used in the object-oriented model to define shop geometries

3.2 Case study selection

A single case study was used for data collection and for the application of the object-oriented model (Fellows & Liu, 2021; Weththasinghe et al., 2022a; Yin, 2017). The chosen case study, situated 27 km west of Melbourne's central business district in Tarneit, is a single-storey subregional shopping centre in climate zone 6 with mild temperatures (Australian Building Codes Board, 2019). It represents the majority of average Australian shopping centres based on gross lettable area (Property Council of Australia, 2023). Currently occupying the largest share of Australian shopping spaces at 5,246,278 m² (constituting 21% of the shopping centre floor space), these single-storey subregional shopping centres have a significant pipeline of projects scheduled for the next few years (Property Council of Australia, 2023; Shopping Centre Council of Australia, 2023). Given that such centres comprise over 80% of their total number, they were chosen as the focal point of this study. The selection criteria were primarily guided by industry benchmarks, including gross lettable area distributions, the number of anchor tenants, the count of specialty shops, and the overall tenant mix. Appendix 1 provides a comprehensive list of benchmark values against which the selected case study was assessed, derived from an analysis of all subregional shopping centres in Victoria (Property Council of Australia, 2023). The internal layout of these centres, specifically the arrangement of specialty shops, adhered to the typical tenant mix stipulated by the Shopping Centre Council of Australia. In the selection process, the statistical parameter 'median' was favoured over 'mean' for its robustness against outliers and ability to offer a more realistic representation of the data set, especially in the presence of a skewed distribution. The case study was used for setting the business as usual (BAU) scenario of the Australian shopping centres in terms of material selection and to assess their life cycle embodied greenhouse gas emissions (LCEGHGE), life cycle material cost (LCMC) and life cycle cost of embodied emissions (LCCEE) incorporating a carbon tax as a measurement.

3.3 Quantification of life cycle embodied greenhouse gas emissions, material cost and cost of embodied emissions

3.3.1 Creating bills of quantities

In order to estimate the life cycle embodied GHG emissions and material costs of different shop types, the model first creates bills of quantities (BOQ) for the shops defined in the *Shops* database. Bills of quantities are created using the basic geometry of parametric shop designs, as discussed in Section 3.1. This process is critical since the quantification of building elements can affect the findings of the study. Furthermore, this automated BOQ generation enables rapid quantification and adds flexibility to the model. Any modifications to the parametric shops in the *Shops* database can easily be adjusted in the *Shop* objects through the automated BOQ generation. The calculation of the quantities of building elements, which are considered as assembly types, is undertaken using the equations presented in Table 2.

Table 2: Equations used to generate bills of quantities of shops in the model

3.3.2 Quantifying embodied greenhouse gas emissions and material cost

In the realm of construction materials, the estimation of greenhouse gas (GHG) emissions can be approached through three methods of life cycle inventory analysis: process analysis, input-output analysis, and hybrid analysis. Previous studies have indicated that the input-output-based hybrid analysis method provides the most comprehensive assessment of embodied flow intensities, offering extensive coverage of system boundaries at the material level (Crawford et al., 2018; Crawford et al., 2022; Stephan et al., 2019). Consequently, this research employs the material embodied GHG emissions coefficients from the EPiC database (Crawford et al., 2019; Crawford et al., 2022), utilising an input-output-based hybrid analysis approach to quantify the LCEGHGE of shopping centres. These coefficients have been specifically tailored to align with the building materials commonly used in Australia, thereby enhancing their contextual relevance. The selection of the EPiC database for this study was based on its recent development and its status as the sole hybrid embodied GHG coefficient database worldwide for construction materials, taking into account the significance of considering the influence of time on material energy coefficients and the reliability of data (Crawford et al., 2019; Crawford et al., 2022).

To assess the LCEGHGE of shopping centres, the calculation of the initial embodied GHG emissions (IEGHGE) and recurrent embodied GHG emissions (REGHGE) of materials and assemblies used in these centres was conducted based on the hybrid GHG emissions coefficients mentioned earlier. When precise emissions coefficient values were not available for certain materials, approximate values were assigned by considering similar materials (Stephan et al., 2012). The REGHGE was determined by taking into consideration the service life, durability, and maintenance requirements of the materials and assemblies. This was accomplished using the replacement rate of assembly equation provided in Table 3. The determination of the replacement rate of materials and assemblies relied on the assembly's service life and the refurbishment frequency associated with different types of shops. These calculations play a critical role in the evaluation of LCEGHGE and material cost figures over the entire lifespan of a shopping centre, thereby influencing the assessment of LCMC.

Calculations of LCMC consists of capital cost (CC) and cost-in-use (CIU) are conducted using the equations presented in Table 3.

Table 3: Equations used in the model to quantify embodied emissions and cost of different scenarios

The evaluation of the impact of a carbon tax is reflected in the LCCEE which measures the cost of embodied emissions of the shopping centres. LCCEE is considered as an aggregation of total capital cost (TCC), which includes the CC and the cost of IEGHGE, and total cost-in-use (TCIU) including CIU and the cost of REGHGE. The LCCEE is then added to the LCMC of respective assembly combinations to estimate the total life cycle material cost (TLCMC). For quantification purposes, the carbon tax value in 2023 was set at AUD 38.00 per tonne CO_2e, based on historical trends of carbon taxes in Australia. It is worth noting that the carbon tax scheme was introduced in Australia in 2012 but was repealed in 2014. The clean energy regulator reported a carbon price of AUD 24.15 per tonne of CO_{2e} emissions for the 2013-2014 financial year (Maryniak et al., 2019). This value was incorporated into the model, considering an annual escalation rate of 4.1%. For the financial year 2022-2023, the estimated carbon price was AUD 38.00. This value was utilised by the model to calculate the total LCMC variable with the inclusion of a carbon tax, accounting for an annual real price escalation rate of 4.1%. To ensure more reliable results, adjustments were made to the model for the year 2023. Given the fluctuating nature of the carbon tax, future values were determined based on a real price escalation rate linked to the inflation rate of goods and services, and then discounted to their present value using the equation presented in Table 3. It was assumed that the real price escalation rate for the carbon tax aligns with the inflation rate. In summary, four key performance indicators are used to assess a shopping centre, namely its life cycle embodied greenhouse gas emissions, life cycle material cost, life cycle cost of embodied emissions, and its total life cycle material cost.

3.4 Scenario Analysis

As described in Section 3.1, various types of assemblies are combined to form a shop. Utilising the assemblies outlined in the *Assemblies* database, diverse combinations are generated for each shop type. These assembly permutations for shop variations serve as the basis for identifying combinations that minimise life cycle embodied greenhouse gas emissions (LCEGHGE), life cycle material cost (LCMC), and total life cycle material cost (TLCMC). The analysis is conducted at the shop level through scenario modelling, encompassing four distinct scenarios: minimum life cycle embodied GHG emissions (LCEGHGE), minimum life cycle material cost (LCMC), minimum total life cycle material cost (TLCMC) and optimal. The optimal scenario represents a situation where minimising LCEGHGE and TLCMC are prioritised equally. These scenarios are then compared to the business-as-usual (BAU) scenario, denoting the most common assembly combination employed in construction and defined in the *Shops_catalogue* database. The analysis considers over 8,000 material combinations to comprehensively explore the various possibilities.

3.5 Sensitivity Analysis

Scenario analyses will be undertaken across different carbon tax scenarios to discern the impact of fluctuations in carbon tax on the decision-making process concerning material selection for Australian shopping centres.

3.6 Data availability

Ensuring data transparency and the reproducibility of results constitutes crucial elements in any research inquiry. In alignment with best practice guidelines from the field of Industrial Ecology (Hertwich et al., 2018), all pertinent supporting data can be accessed through Figshare[1] presented as a citable document.

These data include:

- The database of assemblies;
- The embodied flow and material cost intensities of different shop categories in an Australian shopping centre; and
- Other relevant data.

4 RESULTS AND ANALYSIS

This section presents the outcomes generated through the parametric model in the analysis of the case study shopping centre. The examination focuses on the shop level, where various scenarios are compared to pinpoint assembly combinations characterised by the lowest embodied greenhouse gas (GHG) emissions and material costs, accounting for a carbon tax.

Three primary shop types are exemplified in this section, serving as representations of the typical refurbishment frequencies observed in an average shopping centre, namely the centre structure, anchor shops, and specialty shops. The centre structure encompasses foundational elements, columns, roof structure, structural walls, structural wall finishes, external windows, and lintels. Specialty shops within the model are configured with internal walls, ceiling

[1] https://doi.org/10.6084/m9.figshare.19930022.v1

finishes, wall finishes, and floor finishes. Anchor shops are designed with structural walls, ceiling finishes, wall finishes, and floor finishes. The assembly combinations for these shop variations are utilised to ascertain combinations that minimise life cycle embodied GHG emissions (LCEGHGE), life cycle material cost (LCMC), and total life cycle material cost (TLCMC), contrasting them with the business-as-usual (BAU) scenario—representing the most typical assembly combination in construction—outlined in the *Shops_catalogue* database.

4.1 Analysis of the centre structure

In the model, the centre structure, identified as the largest shop type in the shopping centre, refers to the building shell and is designated as such for modelling purposes.

Table 4: Centre structure details

The discussion delves into the five scenarios, highlighting disparities in assembly selection that lead to variations in embodied energy and material cost. While the refurbishment frequency of the centre structure aligns with the analysis period of the building (set at 50 years), considerations are also given to the service life values of assemblies when determining the frequency of replacements. This approach is taken to quantify recurrent embodied greenhouse gas emissions (REGHGE) and total cost-in-use (TCIU) during the use phase. Consequently, numerous envelope and finishes assemblies may undergo replacements over the analysis period based on their respective service life values, even if the shop itself does not undergo refurbishment. The model compared approximately 720 material combinations, encompassing variations including business as usual (BAU) scenario, which features a concrete slab-on-grade foundation with general-purpose Portland cement, steel columns, a steel roof structure with Colourbond roof sheets, precast concrete panel walls, metal-framed glass windows, steel lintels, and a sheet metal cladding structural wall finish. Table 5 provides a comprehensive overview of the performance of the scenarios pertaining to the centre structure.

Table 5: Comparison of the key performance indicators of the centre structure for different scenarios

Figure 3 illustrates that the initial embodied greenhouse gas emissions (IEGHGE) constitute a predominant proportion of the total life cycle embodied greenhouse gas emissions (LCEGHGE), accounting for 92% - 93% across all scenarios. This prevalence is attributed to the assumption that the centre structure undergoes refurbishment every 50 years, and the service life values of most structural assemblies within the centre structure typically surpass this refurbishment frequency. Consequently, this results in diminished recurrent embodied greenhouse gas emissions (REGHGE).

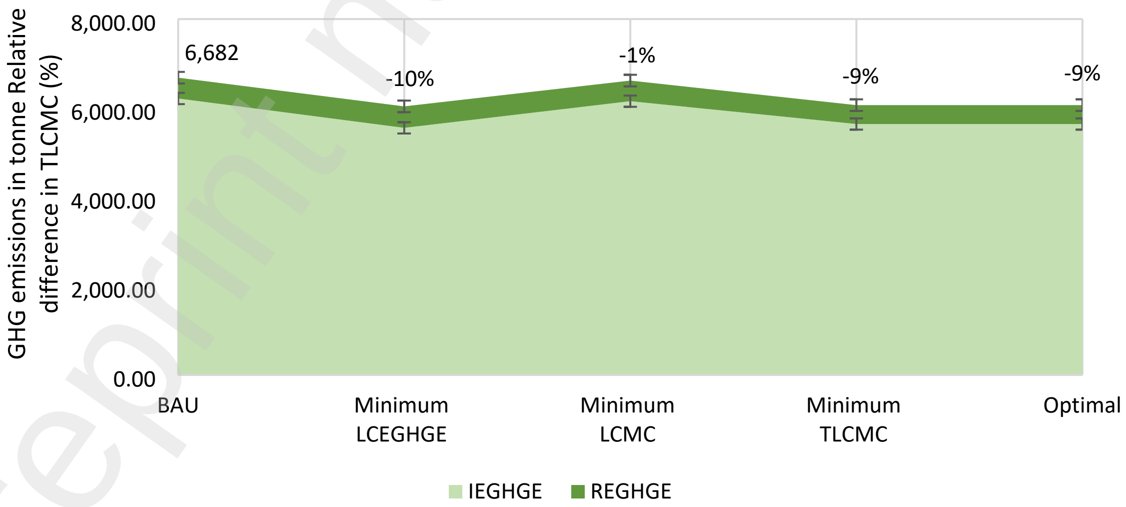

Figure 3: Life cycle embodied GHG emissions comparison of the centre structure across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; IEGHGE: Initial embodied GHG emissions; REGHGE: Recurrent embodied GHG emissions

A parallel observation can be drawn regarding the contribution of the total capital cost (TCC: CC and cost of IEGHGE), which accounts for approximately 98% - 99% of the total life cycle material cost (TLCMC) across all scenarios, as depicted in Figure 4. Total cost-in-use (TCIU: CIU and cost of REGHGE) is only responsible for about 1% - 2% of the TLCMC. Analysis further revealed that changes to material selection could only achieve marginal reductions (up to 1%) to TLCMC.

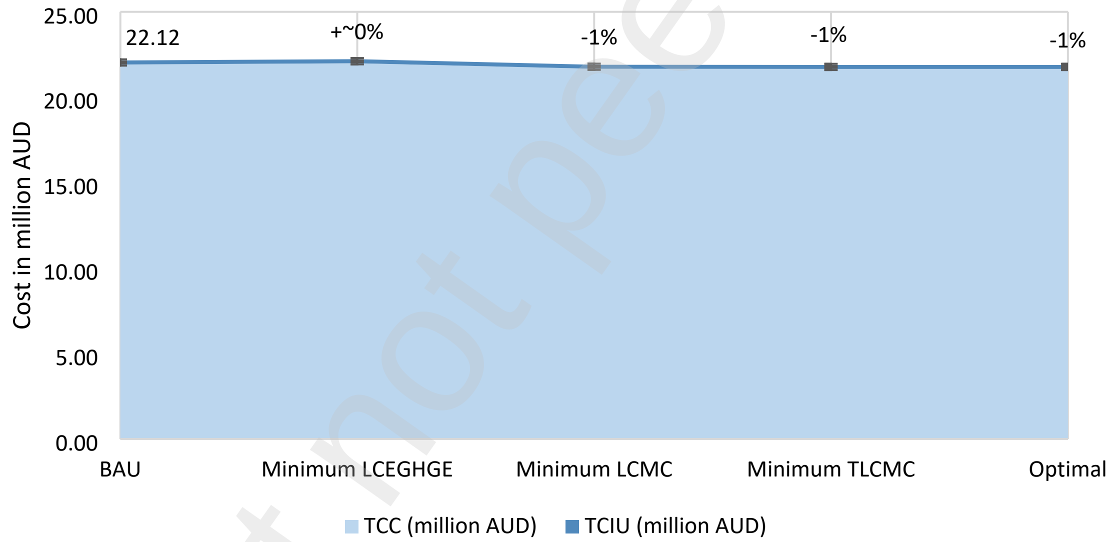

Figure 4: Total life cycle material cost comparison of the centre structure across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; TCC: Total capital cost; TCIU: Total cost-in-use

The BAU scenario represents a typical centre structure design in Australia and is associated with an IEGHGE of 294 kg CO_2e/m^2, while the TCC intensity is 1,036 AUD/m². Notably, the

TCIU and REGHGE for the centre structure are markedly lower in comparison to IEGHGE and TCC. This discrepancy arises from the fact that, over the analysis period, only finishes assemblies undergo replacement. Consequently, the REGHGE is measured at 22 kg $_{CO2e}$/m², and the TCIU is 11 AUD/m². As a result, the LCEGHGE and TLCMC intensities for the centre structure are quantified at 316 kg $_{CO2e}$/m² and 1,047 AUD/m², respectively. However, when the carbon tax is removed from the cost contribution, it is reduced to 1,037 AUD/m². These values are predominantly influenced by the initial assembly input quantities, emphasising the significance of upfront considerations in determining the environmental and economic impacts of building structures.

In comparison to the BAU scenario, the scenario with the minimum LCEGHGE showcases a notable reduction (-10%) in LCEGHGE to 286 kg $_{CO2e}$/m², while the scenario with the minimum LCMC sees a slight decrease (-1%) in TLCMC to 1,035 AUD/m². Interestingly, the scenario with the minimum TLCMC achieves a balance between minimising both LCEGHGE (287 kg $_{CO2e}$/m²) and TLCMC (1,035 AUD/m²). Another vital observation from the analysis is that the optimal scenario whether minimising both LEGHGE and TLCMC are equally prioritised, is similar to the minimum TLCMC scenario. This highlights the potential for optimising both environmental impact and material costs in building design and underscores the significance of employing a carbon tax approach to guide decision-makers towards more environmentally responsive material selection The findings, however, suggest that, in the context of the shopping centre structure, there is limited variability and room for enhancements concerning material selection, indicating the importance of careful consideration in achieving optimal outcomes.

An in-depth examination was carried out to evaluate the initial embodied GHG emissions (IEGHGE) and total capital cost (TCC) within the BAU scenario at the assembly level, taking into account a carbon tax rate of 38.00 AUD/tonne $_{CO2e}$.

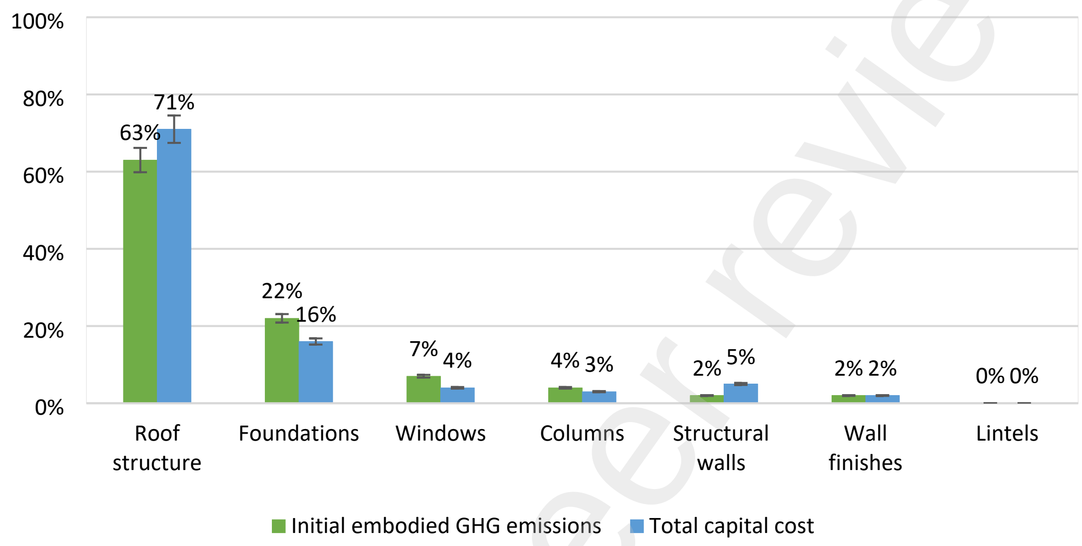

Figure 5: Initial embodied GHG emissions and total capital cost percentage distribution across assembly types in the centre structure business as usual scenario

As demonstrated in Figure 5, the findings reveal that the roof structure plays a pivotal role, contributing to 63% of the total IEGHGE, followed by foundations (22%). In contrast, other assembly types exhibit significantly lower IEGHGE contributions, collectively responsible for around 15% of the total. Examining the TCC contributions yields a similar pattern, with the roof structure being the primary contributor at 71%, succeeded by the foundation at 16%, and the structural wall at 5%. However, it must be noted that these outcomes are closely tied to the quantities of assemblies utilised in the centre structure. This analysis provides insights into the distribution of embodied emissions costs across various assemblies in the BAU scenario, shedding light on key contributors and informing potential mitigation strategies.

4.2 Analysis of the anchor shop

The analysis of anchor shops focuses specifically on a discount department store, which serves as a representative example within the shop category shown in Table 6.

Table 6: Anchor shop details

Table 7 provides a comprehensive overview of the scenarios pertaining to the anchor shop. The comparison involved 149 variations, encompassing the BAU scenario with precast concrete panel walls, plasterboard lining with paint on timber frame ceiling, porcelain floor tiles, and a water-based paint with white putty on cement mortar screed wall finish.

Table 7: Scenario comparison of the anchor shop

The analysis further identifies that changes to material selection could lead to GHG emissions reductions of up to 57% and cost savings of up to 26% when compared with the BAU scenario. The most significant observation here is that the selection of materials in both minimum TLCMC and the optimal scenarios are the same.

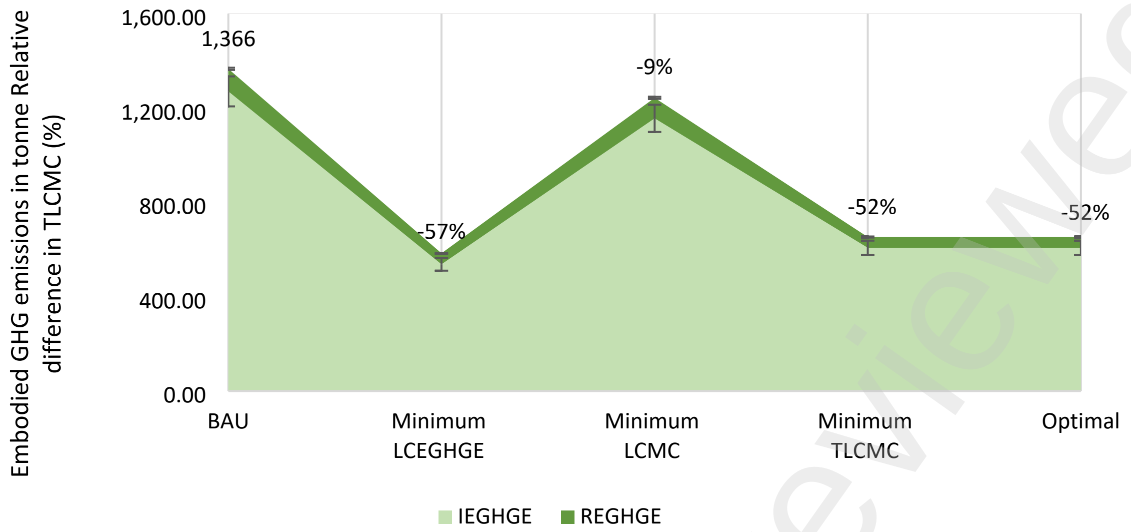

Figure 6: Life cycle embodied GHG emissions comparison of the anchor shop across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; IEGHGE: Initial embodied GHG emissions; REGHGE: Recurrent embodied GHG emissions

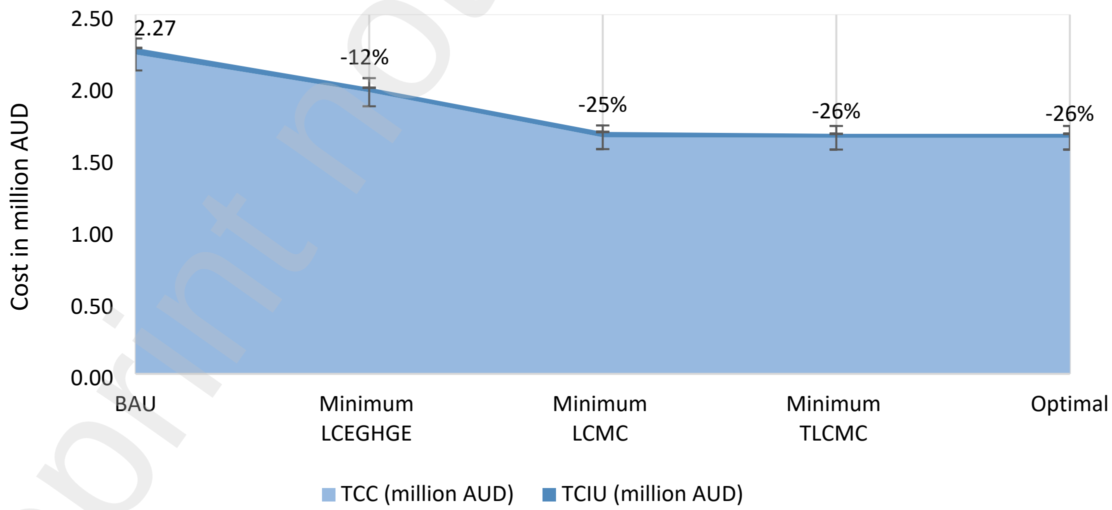

Figure 7: Total life cycle material cost comparison of the anchor shop across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; TCC: Total capital cost; TCIU: Total cost-in-use

Figures 6 and 7 demonstrates that IEGHGE accounts for 92% - 93% of LCEGHGE across all scenarios and the TCC account for 99% of TLCMC in anchor shops. The minimum LCEGHGE scenario has an emissions intensity of 108 kg $_{CO2e}$/m² and total cost intensity of 421 AUD/m² against the minimum TLCMC scenario with 121 kg $_{CO2e}$/m² emissions intensity and cost of 310 AUD/m². The minimum LCMC scenario before carbon tax has an emissions intensity of 230 kg $_{CO2e}$/m², still a reduction of 9% in comparison to the BAU scenario. The optimal scenario is similar to the minimum TLCMC scenario resulting in 52% emissions reduction and 26% total cost saving. The TLCMC across scenarios emphasise the significance of a carbon tax scheme to direct decision makers towards environmentally responsive material selection along with better informed regulatory frameworks enforced.

4.3 Analysis of the specialty shop

Specialty shop analysis is conducted using Services shop type since they represent the largest share of GLA of all specialty shops in the case study subregional shopping centre. Table 8 presents the details of the selected Services shop for analysis of different scenarios. The analysis considered 1,175 variations including the BAU scenario of steel stud wall, water-based paint on plasterboard, timber framed plasterboard ceiling and terracotta tiled flooring.

Table 8: Specialty shop details

Table 9 provides a comprehensive overview of the scenarios pertaining to the specialty shop and shows that across all scenarios, both LCEGHGE and TLCMC are lower compared to the BAU scenario. Behavioural changes in material selection process could lead up to 59% embodied emissions reductions together with up to 48% cost savings. Similar to anchor shops, minimum TLCMC and optimal scenarios are the same in specialty shops. Optimal scenario balances both emissions (51% reduction) and cost (48% reduction) in comparison to the BAU scenario. The analysis at assembly level in BAU scenario further demonstrates that floor finishes (61%) dominate the LCEGHGE followed by ceiling finishes (28%), wall finishes (6%) and internal walls (7%). The TLCMC distribution also takes somewhat a similar pattern led by floor finishes (61%) and ceiling finishes (27%).

Table 9: Scenario comparison of the specialty shop

Examination of the data in Figures 8 and 9 indicate that IEGHGE and REGHGE contributions in services shops are 10% and 90%, respectively. Similarly, TCC only constitutes 14% of TLCMC, while TCIU accounts for 86%. This consistent pattern is observed across all specialty shop types with a refurbishment frequency of five years. Premature assembly replacements every five years are the cause for this pattern as they do not require additional replacements in between, owing to higher expected service life values in all selected assemblies. Consequently, the comparisons stress that, in terms of reducing LCEGHGE in specialty shops, the use phase is more crucial than the initial construction phase. Thus, emphasising material selection during each recurrence becomes imperative for addressing life cycle environmental effects and costs.

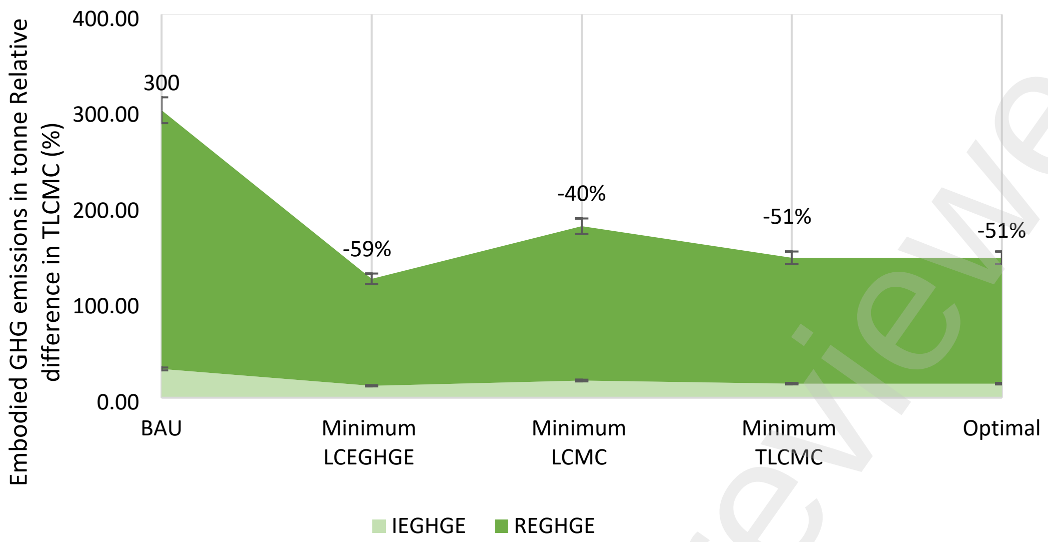

Figure 8: Life cycle embodied GHG emissions comparison of the specialty shop across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; IEGHGE: Initial embodied GHG emissions; REGHGE: Recurrent embodied GHG emissions

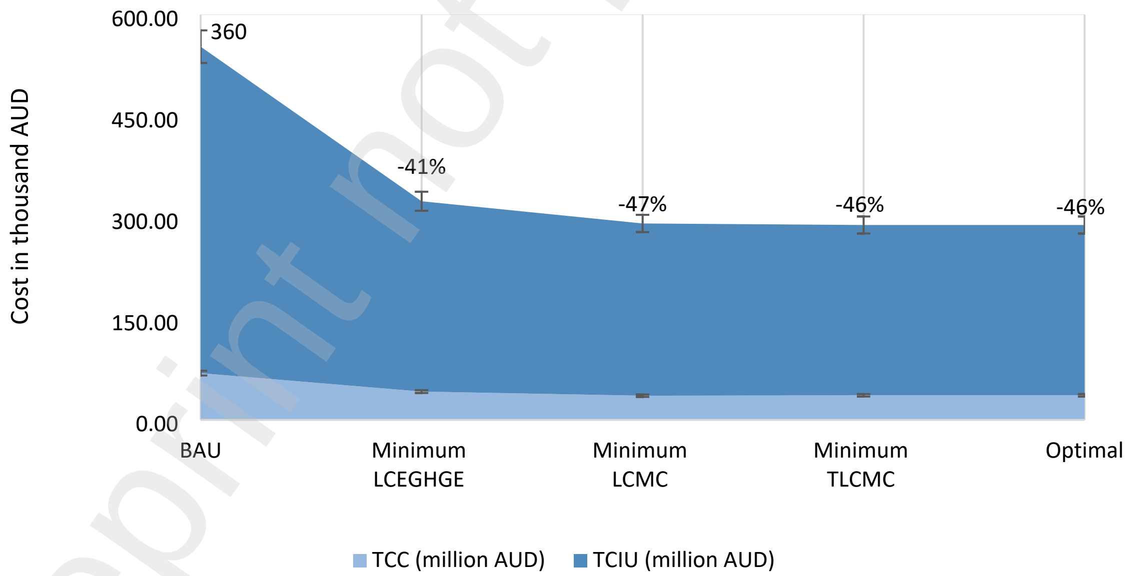

Figure 9: Life cycle material cost comparison of the specialty shop across different scenarios

BAU: Business as usual; LCEGHGE: Life cycle embodied greenhouse gas emissions; LCMC: Life cycle material cost; TLCMC: Total life cycle material cost; TCC: Total capital cost; TCIU: Total cost-in-use

The BAU scenario results in an LCEGHGE intensity of 1,071 $kgCO_2e/m^2$, and TLCMC of 1,973 AUD/m² in comparison to the Minimum LCEGHGE scenario with 423 $kgCO_2e/m^2$ emissions intensity and 1,156 AUD/m² cost intensity. The minimum LCMC scenario where the cost of emissions is disregarded has an emission intensity of 640 kg CO_2e/m^2, marking an emissions reduction of 40%. The analysis reveals that the enforcement of a carbon tax puts significant pressure on material selection decision makers and can be an effective approach for Australia to achieve the established climate change mitigation targets. More legislative frameworks and regulations are also essential to guide decision makers towards environmentally responsive material selection and establish behavioural changes.

4.4 Sensitivity Analysis

This study conducted sensitivity analysis for different refurbishment frequencies (RF) and carbon tax scenarios at shop level to further understand the effects of varying RF and tax values. An analysis was carried out for a specialty shop (services), and the results are presented in Table 10. The business-as-usual scenario RF was set at five years and carbon tax was set at AUD 38.00 and sensitivity was analysed for different values. The BAU scenario was selected for the analysis to understand the optimal RF and carbon tax values that drive decision makers towards the material combinations identified in the optimal scenario of the selected specialty shop category.

Table 10: Sensitivity analysis of refurbishment frequency and carbon tax on life cycle embodied GHG emissions and total life cycle material cost of specialty shops

Figures 10 and 11 below illustrates the relationships between RF and carbon tax with LCEGHGE and TLCMC of the selected specialty shop type across different scenarios.

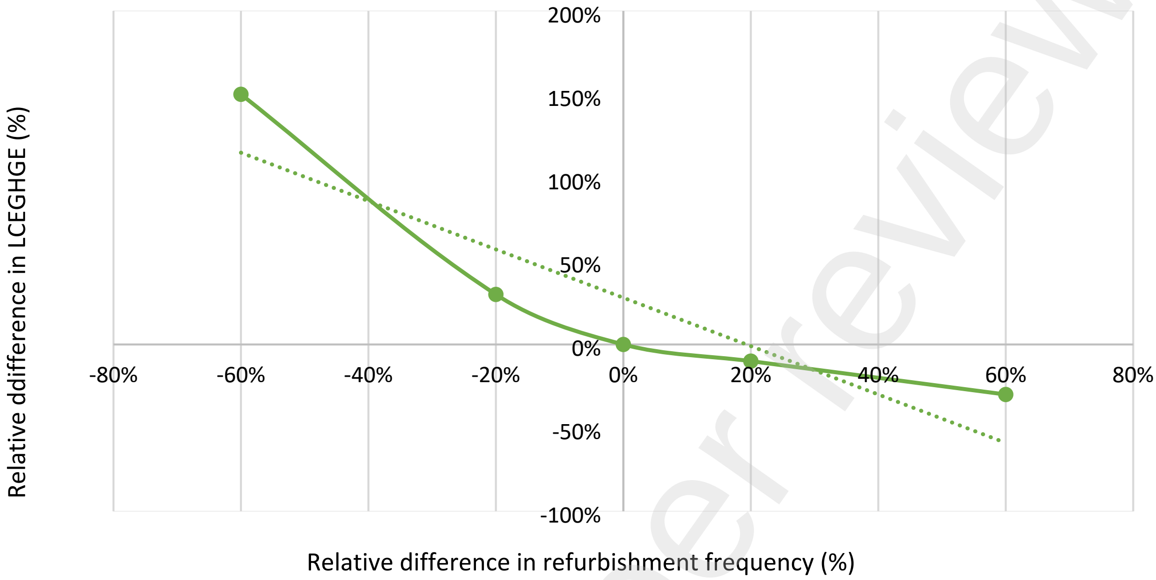

Figure 10: Sensitivity analysis of refurbishment frequency against life cycle embodied GHG emissions in specialty shops

Sensitivity analysis of refurbishment frequency can be approximated by a negative linear relationship with the LCEGHGE of the specialty shop with an R^2 value of 0.8. The greatest reductions in emissions are visible when the refurbishment frequency is increased by 60%. However, it is visible that the increase in emissions is far greater (up to 150%) for lower refurbishment frequencies (-60%) than the reductions achieved (up to -30%) for higher refurbishment frequencies. It is logical that an increased refurbishment frequency will imply less premature material replacements and hence reduce greenhouse gas emissions.

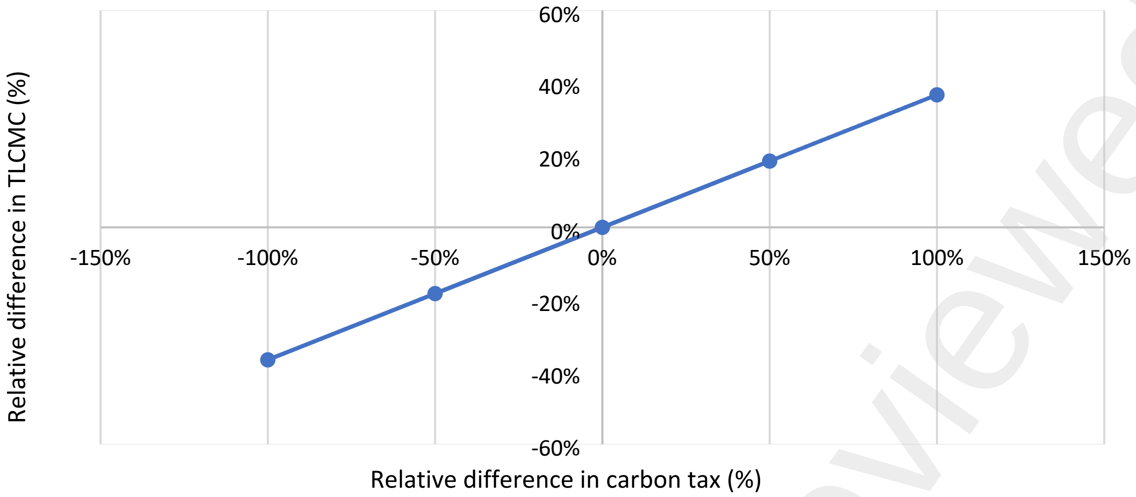

Figure 11: Sensitivity analysis of carbon tax against total life cycle material cost in specialty shops

According to Figure 11, the carbon tax and TLCMC have a positive linear relationship with an R^2 value of 1. An increase in carbon tax by 1% will result in an increase in TLCMC by 0.367%. Results generated from the sensitivity analysis can be used to evaluate how the changes in refurbishment frequency and carbon cost affect the model's outcomes. Those results can be used to make more reliable and robust predictions of the outcome.

5 DISCUSSION

This study uses a Python-based parametric model to examine the implications of implementing a carbon tax on material selection for shopping centres in Australia. The cost of emissions is quantified through life cycle embodied GHG emissions (LCEGHGE), which represents the sum of all emissions associated with building materials and assemblies used in the analysis. The embodied GHG emissions are a crucial measure of global warming and the most pressing sustainability issue globally (Intergovernmental Panel on Climate Change, 2023). Traditionally, embodied GHG emissions have been considered less significant in life cycle assessments compared to operational GHG emissions (Ropo et al., 2023; World Green Building Council, 2019). However, advancements in building materials and technologies have led to the development of more energy-efficient building solutions that can reduce operational GHG emissions (Heydari & Heravi, 2023; Ohene et al., 2022; Stephan et al., 2012). As a result, the relative importance of embodied GHG emissions has increased in recent decades (Akbarnezhad & Xiao, 2017; Gan et al., 2019; Pomponi & Moncaster, 2018), and the assessment of embodied GHG emissions in building architecture and design solutions has become prominent (World Green Building Council, 2019).

The research findings demonstrate that the enforcement of a carbon tax can lead to reductions in embodied GHG emissions of between 9% - 52% in comparison to the BAU scenarios across shop types, while promoting the use of building materials and assemblies with lower embodied greenhouse gas emissions. However, the analysis reveals that the mere introduction of a carbon tax will not result in a shift in material selection compared to the minimum life cycle material cost (LCMC) solutions in some shop types. The business as usual (BAU) scenarios across different shop types usually result in the highest emissions and cost across all scenarios. Since the minimum LCMC scenarios have lower embodied GHG emissions across all shop types, the incorporation of a carbon tax at AUD 38.00 per tonne CO_2e will not necessarily shift the selection decision to more environmentally responsive materials but towards lowest cost options. Importantly, it is observed that the focus should be to identify optimal assembly combinations that minimise both emissions and cost. This finding is a key contribution of the study, as it indicates that the introduction of a carbon tax could indirectly drive stakeholders to select better solutions but only through properly regulated legislative frameworks and policy implementations.

Scenario analysis further revealed that, as the carbon tax increases, there is a tendency to select materials with better environmental performances to reduce the increasing cost of emissions. Thus, the optimisation of the value of a carbon tax is essential to improve environmental performance while considering both environmental and economic aspects. However, implementing a carbon tax is a significant decision with economic implications. In most countries where carbon tax schemes are implemented, they are more of an economy-wide tax (World Bank, 2022). Nevertheless, a more suitable approach could involve developing a construction industry-based tax scheme for assessing the emissions of building materials and assemblies (Laes et al., 2018; Liu et al., 2024; Metcalf, 2019; Wang et al., 2023). Research indicates that carbon taxes directly and significantly impact GHG emissions reduction in a short period (Andersson, 2019; Sadiq et al., 2023; Xu et al., 2023). However, it is crucial to analyse the economic implications before implementing such measures (World Bank, 2022). The addition of a cost component for GHG emissions can lead to increased building materials prices due to cost increments along the supply chain (Sathre & Gustavsson, 2007; Wong et al., 2019; Wong & Zapantis, 2013). Nonetheless, tax swap or tax release schemes can mitigate this economic burden. Many economists argue that carbon taxes are the most efficient and effective means of mitigating GHG emissions with minimal adverse consequences on the economy (Domon et al., 2022; Fang et al., 2024; Gong et al., 2024; Liu et al., 2024).

The carbon tax mechanism was implemented in Australia for two years, from 2012 to 2014 (Burke, 2016). During this period, emissions from the construction industry experienced significant reductions (Copland, 2020; Hammerle et al., 2021). Following the implementation of the carbon tax, the quarterly change in emissions predominantly remained below zero, indicating a reduction in emissions from 2012 to 2014. However, starting from the beginning of 2014, the quarterly change became positive again, indicating an increase in emissions. Although there are some quarters with negative changes (March 2015, June 2016, September 2018), they are not as significant as before. Consequently, it is crucial to promptly evaluate the implementation of the carbon tax scheme in order to guide shopping centre stakeholders toward sustainable designs with lower embodied emissions in Australia.

The results establish that a carbon tax scheme should incentivise the utilisation of building materials and assemblies with lower embodied GHG emissions. Moreover, the government could potentially generate revenue when building materials with higher embodied emissions are used in construction projects and invest those earnings in further emissions reduction strategies. However, unlike other sectors, carbon pricing in the context of the construction of shopping centres needs to account for frequent refurbishments to realistically assess material and assembly options.

These findings provide a solid foundation for policymakers to develop and implement a carbon pricing scheme. Additionally, they offer guidance for shopping centre stakeholders in their pursuit of sustainable designs with emissions reductions in Australia.

5.1 Limitations and further research

This study suffers from limitations related to the assessment of embodied energy, material cost, and greenhouse gas emissions, as well as determining refurbishment frequency and designing shops and shopping centres to reflect reality. The model's quantification processes use a simplified algorithm for replacement rates and material selection, which could be improved by incorporating actual data from a single case study instead of generic rates. End-of-life potential energy recovery and reuse are not included in the assessments.

The shopping centre designs in the model are based on Australian case studies but are simplified representations in terms of use of building materials, tenant mix and the shopping centre layout, potentially affecting embodied emissions and material costs results. The databases include common and energy-efficient materials but lack future, more efficient solutions. Updating databases as new data becomes available could improve the model's accuracy.

Life cycle cost quantification (LCMC and TLCMC) have limitations in labour cost exclusion, material unit price reliability, material replacements, and potential end-of-life cost savings exclusion. Cost data only cover material purchase, transportation, and handling fees, excluding labour and equipment costs for on-site installation. Including labour and equipment costs could impact assembly solutions in terms of minimising embodied emissions and cost. Hence, an initial analysis was conducted to assess the impact of labour costs on the unit costs of internal walls and wall finishes assemblies used in shopping centres. This analysis was a preliminary evaluation of labour intensities, utilising quantitative data sourced from Rawlinsons Quantity Surveyors and Construction Cost Consultants (2023). The comparison revealed that incorporating on-site labour costs resulted in variations in assembly costs. However, the relative ranking of assemblies from least to most expensive remained consistent for all internal wall assemblies, regardless of labour costs. It is worth noting that while the rankings stayed the same, the actual costs of assemblies differed significantly after factoring in labour costs. Nevertheless, this comparison indicated that including on-site labour costs would not alter the model's results at the assembly level. It's important to recognise that labour costs can vary depending on geographic locations, so relying solely on trade ratios may not provide a comprehensive understanding of the situation. Therefore, further research is necessary to fully comprehend how on-site labour costs can influence material selection decisions.

The assessment of TLCMC adopts a straightforward approach by using embodied emissions results and a theoretical carbon tax value specific to Australia. However, this approach is constrained by the limitations inherent in quantifying embodied emissions. Additionally, the carbon tax value utilised is based on 2014 data, and the algorithm incorporates a real price escalation rate for carbon pricing to adjust for expected price increases over the next 50 years. Furthermore, TLCMC values were discounted to present values using the net present value method discussed in Section 3.3.2. The algorithms employed include a discount rate and a real price escalation rate to reflect the time value of money concerning assembly solutions throughout their life cycles. The real price escalation rate is designed to accommodate the anticipated rise in unit prices of materials over time. However, this rate is based on historical financial data and represents a generalised forecasted growth, not specific to individual building trades. This limitation could be addressed by creating a database of price fluctuations for various building trades over the past 50 years. Utilising machine learning or other forecasting algorithms could generate more accurate real price escalation rates for different assembly categories, resulting in a more realistic representation of future financial aspects of building assembly solutions.

The mathematical model heavily relies on various databases for input, with uncertainty managed through interval analysis. Case studies represent typical subregional shopping centres in Victoria, but expanding to other types and locations could enhance the model's applicability and robustness. Incorporating more case studies would also enrich the *Materials* and *Assemblies* databases.

The study acknowledges certain limitations related to model development, particularly concerning data needs and fundamental quantification algorithms. However, the primary focus of this research was to highlight the importance of life cycle embodied impacts and the influence of a carbon tax on material selection decisions in shopping centres. In this regard, the study successfully established a strong foundation for future comparative evaluations of carbon tax scenarios.

6 CONCLUSION AND POLICY IMPLICATIONS

The purpose of this research was to assess life cycle embodied GHG emissions (LCEGHGE) of Australian shopping centres and to evaluate the effect of carbon tax on material selection decision making. Subregional shopping centres were selected for investigation as they are the fastest growing and the largest planned retail floor space in Australia. The study used object-oriented programming (OOP) in Python to implement a mathematical model that simulates different scenarios of building materials and assemblies for shopping centres that optimise different objective functions of minimising emissions and/or cost.

The findings of this study contribute to the body of knowledge and practice and present cost-effective opportunities to achieve emissions reductions for the shopping centres as a built environment asset. The understanding of the effect of carbon tax schemes on material selection decision making is significant in implementing policies and regulations in the near future to mitigate climate change. However, the application of these findings and implementation in the shopping centre industry needs to engage all stakeholders. Primarily, behavioural changes are required from the developers and designers involved in the project inception and design stages to consider the feasibility of using those solutions in the designs as value-added options and to educate clients and other project stakeholders. Furthermore, the shopping centre management needs to be more involved in the design processes of individual shops and to monitor the shop owners' decision making on the selection of building materials and assemblies to guide them towards low emissions alternatives with lower costs. Green star retail interior fit-out rating tool could be used by the centre management as a guide to assess the environmental sustainability of the shop fit-outs. The involvement of the shopping centre management in retail tenants' business can be justified since the implications of individual shops ultimately have a significant effect on the adverse environmental performances of the shopping centre. Also, it is vital to educate shop owners regarding the environmental implications of the shops and the necessity to mitigate them. The low embodied emissions options identified by the model with lower costs can then be appropriately engaged in the shopping centre construction process and throughout the life cycle.

Results are valuable to policy enablers who want to develop specific policies and guidelines dedicated to shopping centres as stipulated below.

- The understanding of the impact of introducing a carbon tax scheme on material selection decisions is crucial for developing effective policies and regulations to address climate change. The study highlights that as carbon taxes increase, there is a tendency to choose materials with lower embodied emissions, although this may lead to economic challenges due to increased costs across the supply chain. Thus, optimising carbon tax policies is essential to promote sustainability while balancing environmental and economic considerations.
- The Green Building Council of Australia (GBCA) can integrate the proposed solutions as preferred materials and assemblies for shopping centres to achieve green star ratings leading to a low embodied emissions material selection and environmental performance. The use of more environmentally responsive materials is already a criterion that is assessed to reward Green Star points in the rating system (Green Building Council Australia, 2022). Hence, this study builds on the existing framework and provides more detailed knowledge to enhance it.
- The findings of this study are useful to GBCA to lay the groundwork for assessing embodied emissions in future projects and finding cost-effective ways to reduce emissions in retail centre construction.
- The benchmarks established in this study are crucial for evaluating the environmental performance of shopping centres. These benchmarks align with existing regulations in Australia's residential building sector (Sustainability Victoria, 2023), suggesting that extending them to the retail sector, based on this study's findings, could gain support. By implementing a policy agenda, regulations akin to those in the residential sector could be applied to shopping centre development, thereby enhancing sustainability standards.

- The insights of assembly combinations and their effects on emissions and costs could inform the creation of policies for sustainable retail fit-out designs. It is essential to differentiate between shop types with different lease periods and refurbishment frequencies, as they require unique design solutions that can significantly reduce embodied emissions without substantial cost increases. These findings highlight the potential for tailored guidelines to achieve emissions reductions in retail fit-out designs while minimising additional costs.

The areas identified in this research present opportunities to improve the environmental performance of Australian shopping centre designs and achieve sustainability goals as a nation.

AUTHOR'S CONTRIBUTIONS

KW: Conceptualisation, Methodology, Data collection, Model development, Analysis, Writing - original draft. PT: Conceptualisation, Supervision, Methodology, Writing - review & editing. VF & AS: Supervision, Methodology, Writing - review & editing.

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REFERENCES

Abdi, A., Taghipour, S., & Khamooshi, H. (2018). A model to control environmental performance of project execution process based on greenhouse gas emissions using earned value management. *International Journal of Project Management*, *36*(3), 397-413. https://doi.org/10.1016/j.ijproman.2017.12.003

Abeydeera, L., Mesthrige, J. W., & Samarasinghalage, T. I. (2019). Perception of Embodied Carbon Mitigation Strategies: The Case of Sri Lankan Construction Industry. *Sustainability*, *11*(11), Article 3030. https://doi.org/10.3390/su11113030

Akbarnezhad, A., & Xiao, J. (2017). Estimation and Minimization of Embodied Carbon of Buildings: A Review. *Buildings*, *7*(1), 5. https://www.mdpi.com/2075-5309/7/1/5

Allen, C., Oldfield, P., Teh, S. H., Wiedmann, T., Langdon, S., Yu, M., & Yang, J. (2022). Modelling ambitious climate mitigation pathways for Australia's built environment. *Sustainable Cities and Society*, *77*, 103554. https://doi.org/10.1016/j.scs.2021.103554

Andersson, J. J. (2019). Carbon taxes and CO2 emissions: Sweden as a case study. *American Economic Journal: Economic Policy*, *11*(4), 1-30. https://doi.org/10.1257/pol.20170144

Australian Building Codes Board. (2019). *Climate zone map: Victoria*. Canberra, ACT, Australian Building Codes Board.

Australian Institute of Quantity Surveyors. (2022). *Australian and New Zealand Standard Method of Measurement of Building Works (ANZSMM)*. Master Builders-Australian Institute of Quantity Surveyors.

Australian Sustainable Built Environment Council. (2022). *Unlocking the pathway: Why electrification is the key to net zero buildings*.

Baranzini, A., Van den Bergh, J. C., Carattini, S., Howarth, R. B., Padilla, E., & Roca, J. (2017). Carbon pricing in climate policy: seven reasons, complementary instruments, and political economy considerations. *Wiley Interdisciplinary Reviews: Climate Change*, *8*(4), e462. https://doi.org/10.1002/wcc.462

Berthe, A., Kubursi, A., Arain, M. A., & Janes, A. (2023). The Comparative Analysis of Carbon Pricing Policies on Canadian Northwest Territories' Economy under Different Climate Change Scenarios. *Energies*, *16*(22), Article 7471. https://doi.org/10.3390/en16227471

Boyce, J. K. (2018). Carbon pricing: effectiveness and equity. *Ecological Economics*, *150*, 52-61. https://doi.org/10.1016/j.ecolecon.2018.03.030

Braslavsky, J. H., Wall, J. R., & Reedman, L. J. (2015). Optimal distributed energy resources and the cost of reduced greenhouse gas emissions in a large retail shopping centre. *Applied Energy*, *155*, 120-130. https://doi.org/10.1016/j.apenergy.2015.05.085

Bueno, C., Hauschild, M. Z., Rossignolo, J. A., Ometto, A. R., & Mendes, N. C. (2016). Sensitivity analysis of the use of Life Cycle Impact Assessment methods: a case study on building materials. *Journal of Cleaner Production*, *112*, 2208-2220. https://doi.org/10.1016/j.jclepro.2015.10.006

Burke, P. J. (2016). Undermined by adverse selection: Australia's direct action abatement subsidies. *Economic Papers: A journal of applied economics and policy*, *35*(3), 216-229. https://doi.org/10.1111/1759-3441.12138

Chen, P.-Y., Chen, B.-Y., Tsai, P.-H., & Chen, C.-C. (2015). Evaluating the impacts of a carbon tax on imported forest products—evidence from Taiwan. *Forest Policy and Economics*, *50*, 45-52. https://doi.org/https://doi.org/10.1016/j.forpol.2014.07.012

Copland, S. (2020). Anti-politics and global climate inaction: The case of the Australian carbon tax. *Critical Sociology*, *46*(4-5), 623-641. https://doi.org/10.1177/0896920519870

Crawford, R. H., Bontinck, P.-A., Stephan, A., Wiedmann, T., & Yu, M. (2018). Hybrid life cycle inventory methods–A review. *Journal of Cleaner Production*, *172*, 1273-1288. https://doi.org/10.1016/j.jclepro.2017.10.176

Crawford, R. H., Stephan, A., & Prideaux, F. (2019). *Environmental Performance in Construction (EPiC) Database* The University of Melbourne, Melbourne. https://doi.org/10.26188/5dc228ef98c5a

Crawford, R. H., Stephan, A., & Prideaux, F. (2022). The EPiC database: Hybrid embodied environmental flow coefficients for construction materials. *Resources, Conservation and Recycling*, *180*, 106058. https://doi.org/10.1016/j.resconrec.2021.106058

Crowley, K. (2017). Up and down with climate politics 2013–2016: the repeal of carbon pricing in Australia. *Wiley Interdisciplinary Reviews: Climate Change*, *8*(3), e458. https://doi.org/10.1002/wcc.458

Curmi, L., Weththasinghe, K. K., & Tariq, M. A. U. R. (2022). Global Policy Review on Embodied Flows: Recommendations for Australian Construction Sector. *Sustainability*, *14*(21), 14628. https://doi.org/10.3390/su142114628

Czachorski, M., & Leslie, N. (2009). *Source energy and emission factors for building energy consumption*.

Dias, W., & Pooliyadda, S. (2004). Quality based energy contents and carbon coefficients for building materials: A systems approach. *Energy*, *29*(4), 561-580. https://doi.org/10.1016/j.energy.2003.10.001

Dixit, M. K. (2017). Life cycle embodied energy analysis of residential buildings: A review of literature to investigate embodied energy parameters. *Renewable and Sustainable Energy Reviews*, *79*, 390-413. https://doi.org/10.1016/j.rser.2017.05.051

Dominioni, G. (2022). Pricing carbon effectively: a pathway for higher climate change ambition. *Climate Policy*, *22*(7), 897-905. https://doi.org/10.1080/14693062.2022.2042177

Domon, S., Hirota, M., Kono, T., Managi, S., & Matsuki, Y. (2022). The long-run effects of congestion tolls, carbon tax, and land use regulations on urban CO2 emissions. *Regional Science and Urban Economics*, *92*, 103750. https://doi.org/10.1016/j.regsciurbeco.2021.103750

Elaouzy, Y., & El Fadar, A. (2023). Role of carbon tax in the widespread integration of sustainable strategies in buildings. *Building and Environment*, *234*, 110182. https://doi.org/10.1016/j.buildenv.2023.110182

Fang, X., He, W., Wen, F., An, M., Song, M., Wang, B., & Ramsey, T. S. (2024). Simulation study on the effect of differentiated carbon tax adjustment on CO2 emissions reduction in China from the perspective of carbon footprint. *Journal of Cleaner Production*, *434*, 140071. https://doi.org/10.1016/j.jclepro.2023.140071

Fellows, R. F., & Liu, A. M. (2021). *Research methods for construction*. John Wiley & Sons.

Fenner, A. E., Kibert, C. J., Woo, J., Morque, S., Razkenari, M., Hakim, H., & Lu, X. (2018). The carbon footprint of buildings: A review of methodologies and applications.

Renewable and Sustainable Energy Reviews, *94*, 1142-1152. https://doi.org/10.1016/j.rser.2018.07.012

Finnegan, S., Jones, C., & Sharples, S. (2018). The embodied CO2e of sustainable energy technologies used in buildings: A review article. *Energy and buildings*, *181*, 50-61. https://doi.org/10.1016/j.enbuild.2018.09.037

Gan, V. J. L., Wong, C. L., Tse, K. T., Cheng, J. C. P., Lo, I. M. C., & Chan, C. M. (2019). Parametric modelling and evolutionary optimization for cost-optimal and low-carbon design of high-rise reinforced concrete buildings. *Advanced Engineering Informatics*, *42*, 100962. https://doi.org/10.1016/j.aei.2019.100962

Gong, Y., Li, Y., Liu, J., & Sun, Y. (2024). Overcoming public resistance to carbon taxes: A cost-efficient solution built on a pre-existing reward-based climate policy. *Journal of Environmental Management*, *352*, 120025. https://doi.org/10.1016/j.jenvman.2024.120025

Gorbach, O., Kost, C., & Pickett, C. (2022). Review of internal carbon pricing and the development of a decision process for the identification of promising Internal Pricing Methods for an Organisation. *Renewable and Sustainable Energy Reviews*, *154*, 111745.

Graabak, I., Bakken, B. H., & Feilberg, N. (2014). Zero emission building and conversion factors between electricity consumption and emissions of greenhouse gases in a long term perspective. *Environmental and Climate Technologies*, *13*(1), 12-19. https://doi.org/10.2478/rtuect-2014-0002

Green Building Council Australia. (2022). *Upfront Carbon Emissions calculation guide – interim: Guidance on calculation methods for the Upfront Carbon Emissions and Life Cycle Impacts credits*.

Green, J. F. (2021a). Beyond carbon pricing: Tax reform is climate policy. *Global Policy*, *12*(3), 372-379.

Green, J. F. (2021b). Does carbon pricing reduce emissions? A review of ex-post analyses. *Environmental Research Letters*, *16*(4), 043004.

Gugler, K., Haxhimusa, A., & Liebensteiner, M. (2023). Carbon pricing and emissions: Causal effects of Britain's carbon tax. *Energy Economics*, *121*, 106655. https://doi.org/10.1016/j.eneco.2023.106655

Gursel, A. P., Masanet, E., Horvath, A., & Stadel, A. (2014). Life-cycle inventory analysis of concrete production: A critical review. *Cement and Concrete Composites*, *51*, 38-48. https://doi.org/10.1016/j.cemconcomp.2014.03.005

Hammerle, M., Best, R., & Crosby, P. (2021). Public acceptance of carbon taxes in Australia. *Energy Economics*, *101*, 105420. https://doi.org/10.1016/j.eneco.2021.105420

Hammond, G. P., & Jones, C. I. (2008). Embodied energy and carbon in construction materials. *Proceedings of the Institution of Civil Engineers-Energy*, *161*(2), 87-98. https://doi.org/10.1680/ener.2008.161.2.87

Hao, H. Q., & Yang, X. (2022). China's Carbon Market in the Context of Carbon Neutrality: Legal and Policy Perspectives. *Sustainability*, *14*(18), Article 11399. https://doi.org/10.3390/su141811399

Hertwich, E., Heeren, N., Kuczenski, B., Majeau-Bettez, G., Myers, R. J., Pauliuk, S., Stadler, K., & Lifset, R. (2018). Nullius in Verba1: Advancing Data Transparency in Industrial Ecology. *Journal of Industrial Ecology*, *22*(1), 6-17. https://doi.org/10.1111/jiec.12738

Heydari, M., & Heravi, G. (2023). A BIM-based framework for optimization and assessment of buildings' cost and carbon emissions. *Journal of Building Engineering*, *79*, Article 107762. https://doi.org/10.1016/j.jobe.2023.107762

Holloway, M. L. (2021). Capture and global warming: the technology and regulation debate. In M. L. Holloway (Ed.), *Innovation Dynamics and Policy in the Energy Sector* (pp. 263-282). Academic Press. https://doi.org/10.1016/B978-0-12-823813-4.00012-9

Huang, B., Xing, K., Ness, D., Liao, L., Huang, K., Xie, P., & Huang, J. (2022). Rethinking carbon–neutral built environment: Urban dynamics and scenario analysis. *Energy and buildings*, *255*, 111672. https://doi.org/10.1016/j.enbuild.2021.111672

Huang, L., Liu, Y., Krigsvoll, G., & Johansen, F. (2018). Life cycle assessment and life cycle cost of university dormitories in the southeast China: Case study of the university town of Fuzhou. *Journal of Cleaner Production*, *173*, 151-159. https://doi.org/10.1016/j.jclepro.2017.06.021

Ibn-Mohammed, T., Greenough, R., Taylor, S., Ozawa-Meida, L., & Acquaye, A. (2013). Operational vs. embodied emissions in buildings—A review of current trends. *Energy and buildings*, *66*, 232-245. https://doi.org/10.1016/j.enbuild.2013.07.026

Intergovernmental Panel on Climate Change. (2023). *Summary for Policymakers* (Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Issue.

Laes, E., Mayeres, I., Renders, N., Valkering, P., & Verbeke, S. (2018). How do policies help to increase the uptake of carbon reduction measures in the EU residential sector? Evidence from recent studies. *Renewable and Sustainable Energy Reviews*, *94*, 234-250.

Langston, C., Chan, E. H., & Yung, E. H. (2018). Hybrid input-output analysis of embodied carbon and construction cost differences between new-build and refurbished projects. *Sustainability*, *10*(9), 3229. https://doi.org/10.3390/su10093229

Liu, Y., Zhao, H., & Li, X. (2024). Environmental policy effects of the carbon tax, subsidy, and policy combinations of China's textile industry: Evidence from the DSGE model. *Journal of Cleaner Production*, *439*, 140791. https://doi.org/https://doi.org/10.1016/j.jclepro.2024.140791

Lolli, N., & Hestnes, A. G. (2014). The influence of different electricity-to-emissions conversion factors on the choice of insulation materials. *Energy and buildings*, *85*, 362-373.

López, L.-A., Cadarso, M.-A., Gómez, N., & Tobarra, M.-Á. (2015). Food miles, carbon footprint and global value chains for Spanish agriculture: assessing the impact of a carbon border tax. *Journal of Cleaner Production*, *103*, 423-436. https://doi.org/https://doi.org/10.1016/j.jclepro.2015.01.039

Luo, W., Zhang, Y., Gao, Y., Liu, Y., Shi, C., & Wang, Y. (2021). Life cycle carbon cost of buildings under carbon trading and carbon tax system in China. *Sustainable Cities and Society*, *66*, 102509. https://doi.org/https://doi.org/10.1016/j.scs.2020.102509

Maryniak, P., Trück, S., & Weron, R. (2019). Carbon pricing and electricity markets — The case of the Australian Clean Energy Bill. *Energy Economics*, *79*, 45-58. https://doi.org/https://doi.org/10.1016/j.eneco.2018.06.003

Mehrzad, P., Taghaddos, H., & Tak, A. N. (2023). Environmental-cost framework to investigate impacts of carbon tax policy on material selection for building structures. *Advances in Building Energy Research*, *17*(1), 98-124. https://doi.org/10.1080/17512549.2023.2171486

Meng, S., Siriwardana, M., & McNeill, J. (2013). The environmental and economic impact of the carbon tax in Australia. *Environmental and Resource Economics*, *54*(3), 313-332.

Meng, X., & Yu, Y. (2023). Can renewable energy portfolio standards and carbon tax policies promote carbon emission reduction in China's power industry? *Energy policy*, *174*, 113461. https://doi.org/https://doi.org/10.1016/j.enpol.2023.113461

Metcalf, G. E. (2019). On the economics of a carbon tax for the United States. *Brookings Papers on Economic Activity*, *2019*(1), 405-484.

Mollaei, A., Bachmann, C., & Haas, C. (2023). Assessing the impact of policy tools on building material recovery. *Resources, Conservation and Recycling*, *198*, 107188. https://doi.org/https://doi.org/10.1016/j.resconrec.2023.107188

Nakano, S., & Washizu, A. (2022). A Study on Energy Tax Reform for Carbon Pricing Using an Input-Output Table for the Analysis of a Next-Generation Energy System. *Energies*, *15*(6), Article 2162. https://doi.org/10.3390/en15062162

Narassimhan, E., Gallagher, K. S., Koester, S., & Alejo, J. R. (2018). Carbon pricing in practice: A review of existing emissions trading systems. *Climate Policy*, *18*(8), 967-991.

O'Gorman, M., & Jotzo, F. (2014). Impact of the carbon price on Australia's electricity demand, supply and emissions. *Crawford School of Public Policy, The Australian National University CCEP Working Paper*(1411).

Ohene, E., Chan, A. P., & Darko, A. (2022). Review of global research advances towards net-zero emissions buildings. *Energy and buildings*, *266*, 112142.

Phillips, D. (2018). *Python 3 object-oriented programming: Build robust and maintainable software with object-oriented design patterns in Python 3.8*. Packt Publishing Ltd.

Pomponi, F., & Moncaster, A. (2016). Embodied carbon mitigation and reduction in the built environment–What does the evidence say? *Journal of Environmental Management*, *181*, 687-700.

Pomponi, F., & Moncaster, A. (2018). Scrutinising embodied carbon in buildings: The next performance gap made manifest. *Renewable and Sustainable Energy Reviews*, *81*, 2431-2442. https://doi.org/https://doi.org/10.1016/j.rser.2017.06.049

Property Council of Australia. (2023). *Shopping centre directory 2023*.

Python Software Foundation. (2023). *Python language reference, version 3.0*. Python Software Foundation.

Quesada-Pineda, H., Smith, R., & Berger, G. (2018). Drivers and barriers of cross-laminated timber (Clt) production and Commercialization: a case of study of Western Europe's Clt industry. *BioProducts Business*, 29-38.

Ratanakuakangwan, S., & Morita, H. (2022). An efficient energy planning model optimizing cost, emission, and social impact with different carbon tax scenarios. *Applied Energy*, *325*, 119792. https://doi.org/https://doi.org/10.1016/j.apenergy.2022.119792

Rawlinsons Quantity Surveyors and Construction Cost Consultants. (2023). *Rawlinsons Australian Construction Handbook 2023*. Rawlinsons Publishing. https://books.google.com.au/books?id=g3SEzwEACAAJ

Röck, M., Saade, M. R. M., Balouktsi, M., Rasmussen, F. N., Birgisdottir, H., Frischknecht, R., Habert, G., Lützkendorf, T., & Passer, A. (2020). Embodied GHG emissions of buildings–The hidden challenge for effective climate change mitigation. *Applied Energy*, *258*, 114107.

Rontard, B., & Hernandez, H. R. (2022). Political construction of carbon pricing: Experience from New Zealand emissions trading scheme. *Environmental Development*, *43*, 100727.

Ropo, M., Mustonen, H., Knuutila, M., Luoranen, M., & Kosonen, A. (2023). Considering embodied CO2 emissions and carbon compensation cost in life cycle cost optimization of carbon-neutral building energy systems. *Environmental Impact Assessment Review*, *101*, 107100.

Sadiq, M., Chau, K. Y., Ha, N. T. T., Phan, T. T. H., Ngo, T. Q., & Huy, P. Q. (2023). The impact of green finance, eco-innovation, renewable energy and carbon taxes on CO2 emissions in BRICS countries: Evidence from CS ARDL estimation. *Geoscience Frontiers*, 101689. https://doi.org/https://doi.org/10.1016/j.gsf.2023.101689

Sathre, R., & Gustavsson, L. (2007). Effects of energy and carbon taxes on building material competitiveness. *Energy and buildings*, *39*(4), 488-494.

Saunders, M. N., Lewis, P., & Thornhill, A. (2023). *Research Methods*. Pearson Higher Ed.

Shi, Q., Ren, H., Cai, W., & Gao, J. (2019). How to set the proper level of carbon tax in the context of Chinese construction sector? A CGE analysis. *Journal of Cleaner Production*, *240*, 117955.

Shipworth, D. (2002). A stochastic framework for embodied greenhouse gas emissions modelling of construction materials. *Building Research and Information*, *30*(1), 16-24. https://doi.org/10.1080/09613210110090412

Shopping Centre Council of Australia. (2023). *Industry Information*. Shopping Centre Council of Australia.

Skillington, K., Crawford, R. H., Warren-Myers, G., & Davidson, K. (2022). A review of existing policy for reducing embodied energy and greenhouse gas emissions of buildings. *Energy policy*, *168*, 112920. https://doi.org/10.1016/j.enpol.2022.112920

Stephan, A., & Athanassiadis, A. (2017). Quantifying and mapping embodied environmental requirements of urban building stocks. *Building and Environment*, *114*, 187-202. https://doi.org/10.1016/j.buildenv.2016.11.043

Stephan, A., Crawford, R. H., & Bontinck, P.-A. (2019). A model for streamlining and automating path exchange hybrid life cycle assessment. *The International Journal of Life Cycle Assessment*, *24*, 237-252. https://doi.org/10.1007/s11367-018-1521-1

Stephan, A., Crawford, R. H., & de Myttenaere, K. (2012). Towards a comprehensive life cycle energy analysis framework for residential buildings. *Energy and buildings*, *55*, 592-600. https://doi.org/10.1016/j.enbuild.2012.09.008

Stephan, A., & Stephan, L. (2020). Achieving net zero life cycle primary energy and greenhouse gas emissions apartment buildings in a Mediterranean climate. *Applied Energy*, *280*, 115932. https://doi.org/10.1016/j.apenergy.2020.115932

Sustainability Victoria. (2023, 2 Aug 2023). *7 Star Homes Program – Participate*. Sustainability Victoria. Retrieved 4 Apr 2024 from https://www.sustainability.vic.gov.au/energy-efficiency-and-reducing-emissions/building-or-renovating/7-star-homes-program/participate#:~:text=Sustainability%20Victoria%20is%20working%20with,to%20assist%20with%20upgrade%20costs

Teh, S. H., Wiedmann, T., Crawford, R. H., & Xing, K. (2019). Assessing embodied greenhouse gas emissions in the built environment. *Decarbonising the Built Environment: Charting the Transition*, 119-141. https://doi.org/10.1007/978-981-13-7940-6_7

Treloar, G. J., Love, P. E., & Holt, G. D. (2001). Using national input/output data for embodied energy analysis of individual residential buildings. *Construction Management and Economics*, *19*(1), 49-61. https://doi.org/10.1080/014461901452076

United Nations Climate Change. (2022). *United Nations Climate Change Annual Report 2022*.

Wang, M., Hu, Y., Wang, S., & Dang, C. (2023). The optimal carbon tax mechanism for managing carbon emissions. *Socio-Economic Planning Sciences*, *87*, 101564. https://doi.org/https://doi.org/10.1016/j.seps.2023.101564

Weththasinghe, K. K., Stephan, A., Francis, V., & Tiwari, P. (2020). Use of an object-oriented system for optimizing life cycle embodied energy and life cycle material cost of shopping centres. Sustainability in Energy and Buildings: Proceedings of SEB 2019,

Weththasinghe, K. K., Stephan, A., Francis, V., & Tiwari, P. (2022a). Improving material selection in shopping centres through a parametric life cycle embodied flow and material cost analysis model. *Renewable and Sustainable Energy Reviews*, *165*, 112530. https://doi.org/10.1016/j.rser.2022.112530

Weththasinghe, K. K., Stephan, A., Francis, V., & Tiwari, P. (2022b). *Supplementary information for Improving material selection in shopping centres through a parametric life cycle embodied flow and material cost analysis model* Version 1) figshare. https://doi.org/10.6084/m9.figshare.19930022.v1

Wong, P. S., Holdsworth, S., Crameri, L., & Lindsay, A. (2019). Does carbon accounting have an impact on decision-making in building design? *International Journal of Construction Management*, *19*(2), 149-161. https://doi.org/10.1080/15623599.2017.1401290

Wong, P. S. P., & Zapantis, J. (2013). Driving carbon reduction strategies adoption in the Australian construction sector – The moderating role of organizational culture. *Building and Environment*, *66*, 120-130. https://doi.org/https://doi.org/10.1016/j.buildenv.2013.04.022

World Bank. (2022). *Carbon Pricing Leadership Report 2021/22*.

World Green Building Council. (2019). *Bringing embodied carbon upfront: Coordinated action for the building and construction sector to tackle embodied carbon*.

Xu, H., Pan, X., Li, J., Feng, S., & Guo, S. (2023). Comparing the impacts of carbon tax and carbon emission trading, which regulation is more effective? *Journal of Environmental Management*, *330*, 117156. https://doi.org/https://doi.org/10.1016/j.jenvman.2022.117156

Yin, R. K. (2017). *Case Study Research and Applications: Design and Methods* (6th ed.). SAGE Publications, Inc.

Yu, M., Robati, M., Oldfield, P., Wiedmann, T., Crawford, R., Nezhad, A. A., & Carmichael, D. (2020). The impact of value engineering on embodied greenhouse gas emissions in the built environment: A hybrid life cycle assessment. *Building and Environment*, *168*, 106452. https://doi.org/10.1016/j.buildenv.2019.106452

Yu, M., Wiedmann, T., Crawford, R., & Tait, C. (2017). The carbon footprint of Australia's construction sector. *Procedia engineering*, *180*, 211-220. https://doi.org/10.1016/j.proeng.2017.04.180

APPENDICES

Appendix 1: Case study shopping centre profile against industry benchmark values

Criteria	Median of benchmark values	Case study values	Relative difference
No of anchor tenants	3	3	0%
Anchor tenants-Gross lettable area retail (m²)	11,660.00	12,100.00	+4%
No of specialty stores	49	47	-4%
Specialty- Gross lettable area retail (m²)	6,381.00	5,802.00	-9%
Total Centre- Gross lettable area retail (m²)	17,490.00	20,250.00	+16%
Total Centre- Gross lettable area (m²)	18,426.00	22,498.00	+22%
Specialty proportion	34.44%	39.00%	+13%
Anchor proportion	63.00%	60.00%	-5%
Cinemas	No	No	No difference
Centre type	Enclosed	Enclosed	No difference
Ventilation	Fully airconditioned	Fully airconditioned	No difference
Enclosed car bays	No	No	No difference
Tenant mix			
Anchor			
Supermarket	32.00%	28.15%	-12%
Discount department store	28.00%	31.60%	+13%
Specialty			
Clothing	4.00%	2.00%	-50%

Food supplies	5.50%	7.38%	+34%
Household	5.25%	11.54%	+120%
Multimedia and electronics	1.25%	N/A	N/A
Gymnasium	1.50%	2.01%	+34%
Leisure and entertainment	2.75%	0.46%	-83%
Health and beauty	1.50%	4.80%	+220%
Coffee and restaurant	2.50%	4.71%	+88%
Other retail	2.50%	0.10%	-96%
Shoes	1.25%	0.52%	-58%
Services	12.00%	6.73%	-44%